

UNITED STATES DEPARTMENT OF AGRICULTURE
FOREST SERVICE

R-1

REPLY TO: 2410 Plans

February 27, 1980

SUBJECT: Permanent Growth Plots

TO: Forest Supervisors



Enclosed find instructions for installing permanent growth plots. If further information is needed, contact me at FTS 585-3831 (329-3831).

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PERMANENT GROWTH PLOT INSTALLATION INSTRUCTIONS

As we continue to use models for the construction of yield tables in timber management planning and for the assessment of treatment alternatives in silviculture, it is important to be able to check model accuracy. Inaccurate predictions can only lead to a Forest having inaccurate cutting levels. Gross bias in a model should be intuitively obvious to the user. But what about less obvious bias, such as basal area bias of 5 square feet and height bias of 10 feet. Between height and diameter, a 15 percent total volume bias is easily possible. To monitor this kind of problem, a statistically sound monitoring program is essential. The use of these guidelines can help in setting up statistically sound permanent growth plots. The permanent plots described here consist of at least four cluster plots randomly located in the stand. One of the plots is a control plot with no treatment.

Stand Selection

Because growth plot installation is expensive, species and habitat types most significant to the Forest should be sampled first. Most preferable are stands between 10 and 80 years of age, for young stands are rare in the Region's data base and rare in the data that is in the growth prognosis system. The stands to be monitored should be scheduled for treatment but not yet treated. If stands are already treated, it is nearly impossible to reconstruct the plots to pretreatment conditions. The scheduled treatment must not be a regeneration harvest.

Stands are to be at least 20 acres in size. They need to have homogeneous slope, aspect, habitat, and timber condition class. The elevation change in the stand must not exceed 500 feet.

Kind and Number of Plots

In general, the more complex a stand's structure is, the more plots are needed to achieve a given statistical error. The design described in this paper is an attempt to account for this statistical reality and still keep the installation of plots simple. A second problem with sampling by permanent plots is that the number of trees per plot usually declines with time. In order to get enough trees later on, a lot of trees need to be measured now. This is an unfortunate circumstance which must be tolerated.

As a minimum, four clusters of 3 plots each will be used. If at least 40 trees (3 inches and larger) are not measured per cluster, additional clusters will be added until the total number of measured trees is at least 160.

Each cluster consists of three 1/20-acre fixed radius plots. Small trees are subsampled with three 1/300-acre plots in each 1/20-acre plot.

Locating the Clusters

1. Stand maps - Delineate the stand to be treated on both resource aerial photographs and topographic maps at approximately 1:15,000-1:24,000 scale. Further delineate the stand into general areas of homogeneous slope, aspect, elevation, and timber condition. Identify all roads, probable landing sites, and other areas known to not be available for treatment (stream protection zones, etc.). Select the homogeneous area you are more interested in and outline it for easy reference on both the map and photo.

2. Cluster centers - As noted in figure 1, place an X-Y coordinate over the stand delineated on the topographic map so that the (0, 0) coordinate is in the lower left corner. Randomly select X, Y coordinates using a calculator or the computer program noted in the appendix. Locate four random cluster centers within the stand of interest.

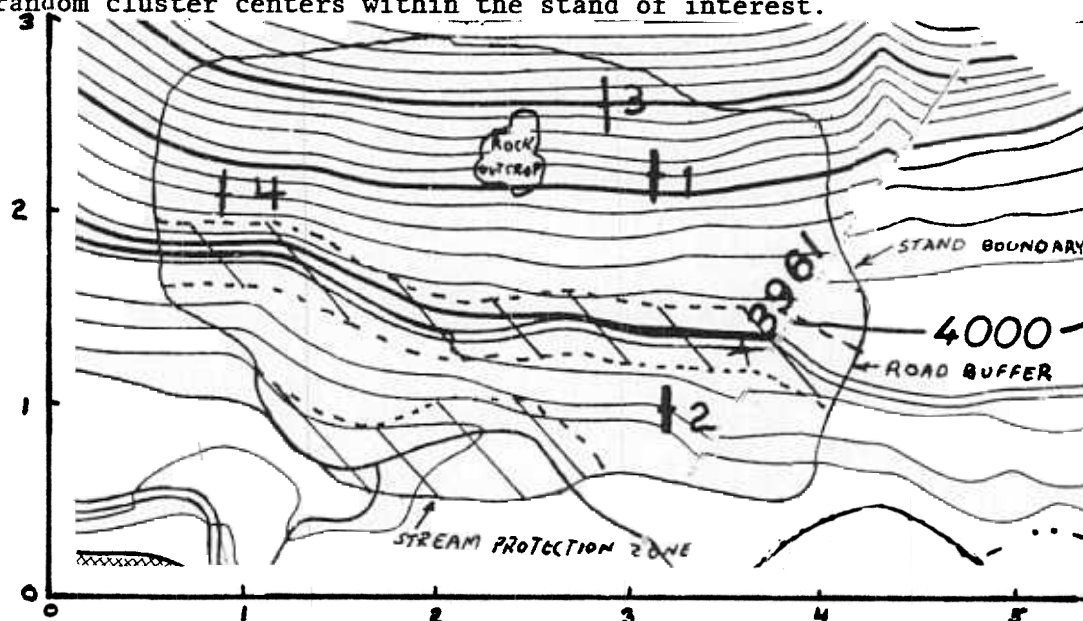


Figure 1. Randomly locating plots on the topographic map

The first cluster will be designated as a control. If any other cluster falls within one chain of the control, delete it and draw another until one is found that is not within one chain of the control plot. Each X-Y coordinate picked is actually the center of a cluster of 3 plots. The plots in a cluster fall along a straight line with 1.5 chains between plot centers. Figure 2 shows the cluster plot makeup.

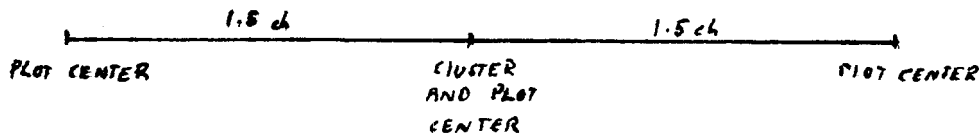


Figure 2. The cluster plot

The azimuth of the line from plot to plot in a cluster need not be a cardinal direction; however, each cluster in the stand will have the same azimuth. The azimuth shall be decided before the plot is located in the field. Decide the azimuth so that each cluster is as heterogeneous as possible. This is to say, run the clusters across skyline corridors, up and down hill or across vegetative fingers. Figure 3 gives some examples.



Figure 3. Make the cluster heterogeneous

Each plot is 1/20-acre in size for trees 3 inches d.b.h. and larger. Within each 1/20-acre plot, small trees are subsampled with three 1/300-acre plots. Small tree plots are located one in the center and two near the 1/20-acre plot boundary on the azimuth line as shown in figure 4.

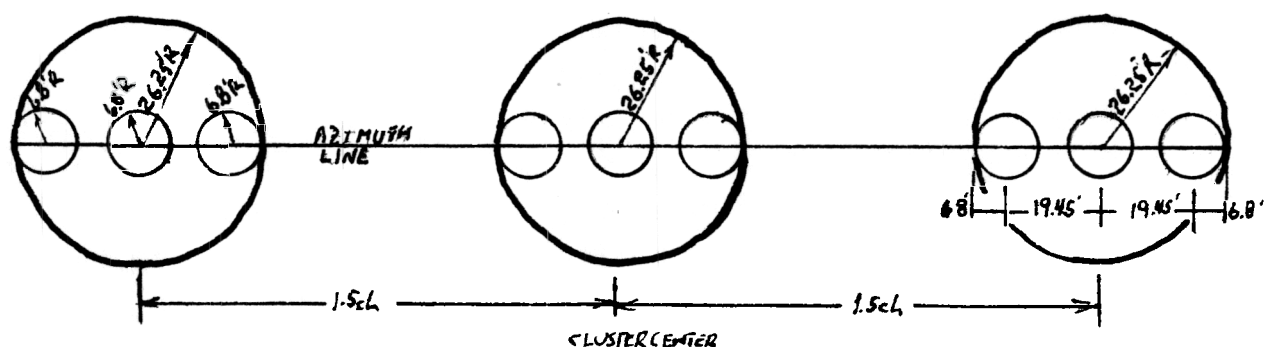
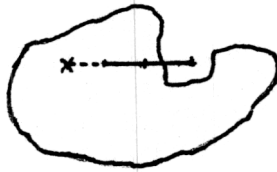


Figure 4. The complete cluster plot

3. What if a plot in a cluster would fall outside the stand? If a cluster falls near a boundary, be sure no plots fall wholly or partially outside the stands. At the risk of some minor bias, place the "outside" plot on the other end of the plot line.

Example:



This plot is part outside; place it where X is. (If that is outside, go 180 degrees and try that.)

1 treatment.

4. Stands under 20 acres - For reasons not to be discussed here, a stand might be selected which is less than 20 acres in size. For best results in small stands, use a systematic sample of 10 plots layed out in a grid fashion. Randomly select 2 plots for control treatment. All other information concerning data to collect, plot size, documentation will be the same as the cluster design used in large stands.

Treatments - For most purposes, only two treatments will be considered. One treatment will be a control (no treatment). One treatment will be that prescribed by the silviculturist. Control plots are as important as the prescribed treatment. A question frequently asked is, "Did the treatment produce a better result than doing nothing?" Going to a nearby "similar" condition is not always a sound way to find a control when statistically significant differences are desired.

Equipment needed for plot installation:

- a. One 6-foot post 2 feet into the ground for each plot (i.e., 3 plots per cluster for 4 clusters would be 12 posts). Metal pins to mark small tree plot centers.
- b. Five hundred metal numbers.
- c. Aluminum nails - at least 2 inches long.
- d. STAND EXAMINATION FIELD RECORD FORMS (R1-2410-15B) (one for each plot in a cluster).
- e. Steel tape

- f. Metal number stamp (for stamping plot number on fencepost).
- g. Tree paint in spray can
- h. Claw hammer, carpenter's apron.
- i. Aerial photos of stand with cluster centers pricked and stand delineated.
- j. Maps of vicinity and stand.
- k. Pressboard folder with Kraft inner divider GSA 7530-00-990-8884 for storing plot records.
- l. D-Tape.
- m. Clinometer.
- n. Relaskop if variable radius plots are used.
- o. Wire pins for marking the center of each small tree plot.

Tree information - (Form R1-2410-15A) For trees 3.0 inches and larger, record and measure all stand exam information except age, height, and diameter increment for all trees. Record age and diameter increment on a random selection of the first tree in each of the following diameter classes for each species. Record heights on all trees on one-half of the plot unless a tree has an unusual height for its diameter. Large trees cannot have numbers attached to them that are greater than 500.

3 - 4.9
5 - 8.9
9 - 12.9
13 - 16.9
17 - 20.9
20.9+

Small tree subplots will be kept track of from the tree number, one subplot will have tree numbers 501-600, one subplot will have tree numbers 601-700, and one subplot will have tree numbers 701-800. These numbers are for recording only as small trees are not tagged in the field. On a sketch map on the field sheet, indicate which numbers are associated with which subplot.

For trees less than 3.0 inches, follow the guidelines in the stand examination handbook for small trees. Do not code trees 0 to 1 inch as .1 d.b.h., as FSH 2409.21 R-1 states. Record the actual diameter. Measure the azimuth and distance to the four best small trees for each 1/300-acre plot; record in the plot sheet margin. Other standards of measurement and coding are the same as shown in the stand examination handbook except:

Transect # = Cluster #

Point # = Plot #

BAF = 83. Make a note next to this that the size is one-twentieth subsampled with three 1/300-acre plots.

Nail and tag all trees that will have a good chance of being left after treatment. Measure d.b.h. after a nail is in the tree on the uphill side. Paint spot all trees after measurement so there is no duplication of records. Tag the tree below stump height facing plot center. Label the plot stakes with the cluster and plot number.

If the plot is a control plot, flag a buffer around the plot boundary one-half chain wide. Flag it well so the buffer will not be treated.

After the cluster is installed, compass and chain the distance from the plot closest to the road, and record it on the back of the plot sheet. Note any helpful information on how to locate the cluster. Place a takeoff point marker on the road. Describe your marker and the kind of tree it is on.

After treatment, the plot will be revisited and a check made of the residual stand. All trees not tagged will be tagged at this time. All tagged trees will be listed and checked on the plot sheets. Subsample plots will also be rechecked.

Documentation - Keep the records for the stand in a pressboard folder. In the folder will be aerial photos, maps, plot sheets, directions on how to get to the stand and clusters. The folder also will contain information on general stand conditions and a copy of the silvicultural prescription. Note information on date of treatment and who did the work and with what equipment. This file will be kept with the Forest coordinator for permanent growth plots. A copy will be sent to the Regional Office Plans and Programs Branch. It is important that all the items mentioned above be in the folder. This means ordering stereo photo pairs for all the file folders. Each pair needs the stand delineated and the plot centers pricked.

APPENDIX

STAND EXAMINATION FIELD RECORD

3- $\frac{1}{2}$ 20m
USUALLY 4

TYPE OF SURVEY		12	ELEVATION	XX	
STATE	XX		SLOPE	X	
COUNTY	XXX		ASPECT	X	
FOREST	XX	89	SLOPE POSITION	X	
BLOCK	X		LOGGED	X	
COMPARTMENT	XX				
SUBCOMPARTMENT	X				
SUBUNIT	X		AREA	XXX	
STAND	XX		PI CLASS	XX	
BAF	XX	83	NO. 1 PRESCRIPTION	XXX	
NO. OF POINTS	XX		NO. 2 PRESCRIPTION	XXX	
HABITAT TYPE	XXX		TIMBER LAND USE CLASS	XXX	

OBSERVATIONS AND REMARKS

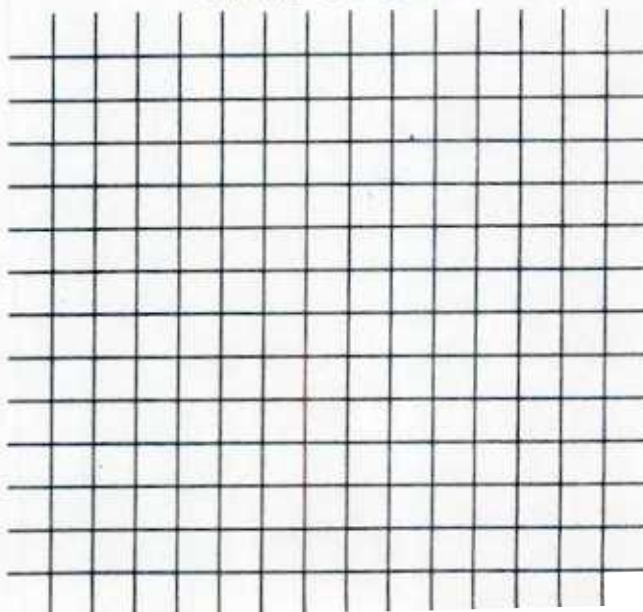
INFORMATION ON HOW TO GET TO THE STAND

TAKE OFF POINT DESCRIPTION - HOW TO GET TO THE CLUSTER

DESCRIBE PLOT CENTER STAKES

OTHER USEFUL INFORMATION ON THE STAND-PLOTS

STAND DIAGRAM



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Chapter 600 - Special Surveys

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620 - PERMANENT GROWTH PLOT SURVEYS. This section describes the purposes, nature, and installation of permanent plots. Targets in terms of installations effected will be negotiated with Forests annually. A 5-year program of installation is envisioned, followed by remeasurement on a 5-year cycle.

621 - Purposes of Permanent Plot Installations

1. To provide a means of assessing and predicting the results of silvicultural treatment with respect to both:

- a. Adherence to the stand's prescription.
- b. Response to treatment.

2. To monitor the Forest plan with respect to:

a. Applicability of the general prescription given by the plan. (This function is also served by stand examination.)

b. Assessing accuracy of yield tables used in the linear programming model for Forest planning.

The growth and mortality information obtained from permanent plots will generally be used by means of a stand simulation model--the Prognosis Model or its successors. Remeasurement data will be used to update and reformulate portions of the present model or develop future models.

621.1 - Assessing and Predicting Results of Silvicultural Treatment. Permanent plots can be used to assess response to:

1. Treatment of a particular condition which occurs on the Forest.

2. Treatment of a particular stand and, by inference, the response to treatment of similar stands.

To illustrate this, suppose certain stands tend to be rather heterogenous with respect to stocking. Suppose regeneration is excessively dense in most parts of such stands, but 1 to 5 acre spots exist which are of no more than adequate or less than adequate stocking. The spots are too small to delineate as separate stands, but the prescription will result in thinning only in the overdense portions of the stand. Now, if it is desired to assess the response to thinning in an overdense condition, then plot clusters, including the control, should be situated only in the overdense portions of the stand. This may require a conditioned random approach to cluster location rather than a purely random approach. This is described under locating

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plot clusters--locating cluster centers. On the other hand, if the objective is to assess application of the prescription and growth response in that particular stand, then plot clusters should be located purely at random, with the control cluster chosen at random.

If the data obtained from permanent plots is used to update a stand model, and to calibrate the model to the stand in some cases, then it matters little whether stands or conditions are sampled by the permanent plots. The data to which the growth model is subsequently applied must be from a representative sample of the stand. Permanent plot installations do not serve the purpose of a stand examination. They should not be used for writing prescriptions or as compartment inventory sample stands.

622 - Selection of Stands and Timing of Plot Measurement

622.1 - Selection of Stands. Because growth plot installation is expensive, species, habitat types, and topographic conditions most significant to the Forest should receive highest priority for permanent plot installation. Significance to the Forest may often be strongly influenced by the Forest plan. A useful approach to this is to make a large table showing in its cells, acreage by a cross-classification of conditions existing on the Forest. Obviously, acreage is important. A highly productive habitat type of only 1,000 acres is a lot less important than 300,000 acres of mediocre productivity. It will become obvious right away that some collapsing of classes will be necessary because enough plot installations to fill all of the cells will be out of the question. In this, be guided by the numbers of acres involved as well as by ecological similarity.

Stands selected for permanent plot installations should be between 10 and 80 years of age. It is between these ages that most stand growth occurs and that intermediate cuttings are feasible. Therefore, we need information on response to treatment during this span of ages. The stands selected for plot installation should be scheduled for treatment in the "very near future," but not yet treated. Very near future means this year or next. Any longer time between plot installation and treatment will cause data taken for the control plots to be out of date if taken at the time of installation. In general, the scheduled treatment should not be a regeneration cut, unless in a particular case it is desired to monitor regeneration establishment and predict future yields under a shelterwood.

In some cases there may not be a sufficient number of suitable stands scheduled for treatment to allow installing all of the permanent plots programmed for a District. In that case, plots can be installed in stands already treated. This will not allow for a control cluster to represent the untreated condition. Nevertheless, one cluster in each stand should be randomly

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selected as a control cluster and set aside from any future treatment in the stand. This practice should be kept to a minimum. If lack of suitable stands scheduled for treatment is a serious problem, consult TM Planning about targets.

If possible, stands should be at least 20 acres in size to avoid problems in locating plot clusters. For the same reason, "stringy" stands or those with lots of fingers are to be avoided. Stands should have homogeneous slope, aspect, habitat, and timber condition class. If this is not possible, then one must decide whether to sample the stand as it exists, or the condition that most of the stand represents. Generally, it will be advisable for permanent plots to sample the condition within the stand that will be treated. If an unbiased sample of the entire stand is desired, a silvicultural stand examination will better serve the purpose. So far as possible, the following criteria should be met:

1. Aspect within the portion of the stand where plots are located should not vary more than one aspect class. For example, a stand may contain both north and northeast aspects, or north and northwest aspects, but not northwest, north, and northeast aspects. Local differences in aspect covering only a small area, such as might be caused by a hummock or dissecting ravine, are not to be considered in this, except that they are to be avoided in locating clusters.
2. Slope within the portion of the stand where the plots are located should not vary more than 20 percent.
3. All of the stand where plots are located must be of the same habitat type.
4. The elevational difference within the portion of the stand where plots are located should not exceed 500 feet.

622.2 - Timing of Plot Installation and Measurement. There are two important time periods in the establishment of permanent plots: the time before the stand is thinned and the time after it is thinned. Selection of the stand and establishment of plot centers with necessary monumentation and documentation will usually be done before the stand is treated. Actual measurement will probably be done after the stand is treated. A pretreatment measurement might have some value, but it would add significantly to the cost of permanent plots. Many trees would be measured that would soon be removed. The plots would also have to be visited after thinning to determine which trees were left to constitute the growing stock.

622.3 - Scheduling Permanent Plot Installation. There are three important considerations in scheduling permanent plot installation and measurement:

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1. The control plot cluster must be established with a 1/2 chain buffer strip around it before the stand is treated. The thinning crew leader should be shown the control cluster location on an aerial photo covering the stand or even on the ground if there is any doubt about where the control cluster is. Mark the buffer strip boundary distinctly with paint and/or flagging.

2. Treated plot clusters must be visited after the stand is thinned, even if trees on those plots were measured earlier. Otherwise, which trees remain as growing stock will remain unknown.

3. Control plots and treated plots should be measured at the same time, or at least at such times that the same growing seasons will elapse between measurements for both sets of plots. Otherwise, growth measurements will not be comparable. If the plots are measured at different times, but between the same two growing seasons, this requirement will be met.

622.4 - Establishment and Measurement of Permanent Plots. It is best if growth sample plots are measured before growth starts in the spring or after growth has effectively ceased in late summer or fall. A suggested sequence for establishment and measurement of permanent plots follows:

1. Select a stand. Locate and monument cluster and plot centers. Flag out a 1/2 chain buffer around the control plot clusters. Record site variables and records of witness trees, route to plots, etc.
2. Thin the stand, making sure the control plot cluster is untouched.
3. Measure all trees remaining on the plots.
4. Complete documentation and plot records.

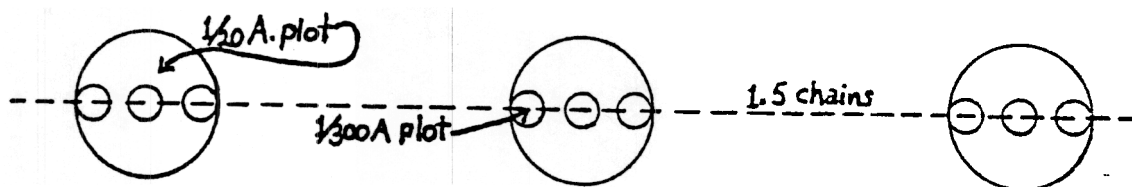


Figure 1. Schematic diagram of a permanent plot cluster.

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In general, the more complex a stand's structure, the more plots are needed to achieve a given statistical error. With permanent plots, we are trying to achieve a sample that is sensitive to the effect of treatment on tree and stand growth rather than to sample the stand itself. The design given here is an attempt to achieve this and still keep the installation of plots simple. One problem with permanent plots is that tree size increases and number of trees per plot decreases with time. In order to have enough sample trees on the plots in 20 or 30 years, the plots must be large enough to contain more trees than necessary at the first few measurements. There is no way to avoid this.

The 1/20 acre size specified for the larger plots is a guideline only, not a hard and fast rule. This plot size is generally appropriate for young stands of the type in which most permanent plots will be installed. Throughout this Chapter, the larger fixed area plot will often be referred to as the 1/20 acre plot for the sake of simplicity. This does not mean that only 1/20 acre plots can be used. If sparse stand density or large tree size should so indicate, plots of 1/10 acre or 1/5 acre can be used. The guiding concept is that each cluster should contain at least 50 trees at the end of the installation's usefulness. This is expected to be at least 30 years in the future. Fifty trees on three 1/20 acre plots is equivalent to 333 trees per acre. If the stand is expected to have significantly fewer trees than this 30 years after plot installation, say fewer than 250 trees per acre, for example, then use of larger plots should be considered. Appropriate codes for entry in the basal area factor field (Field 13) to designate various fixed plot sizes are:

1/20 acre plots	BAF = 83
1/10 acre plots	BAF = 84
1/5 acre plots	BAF = 85

If 1/5 acre plots are used, the distance between plot centers in the cluster must be increased to 2 chains. The fact that this was done must be included in the documentation.

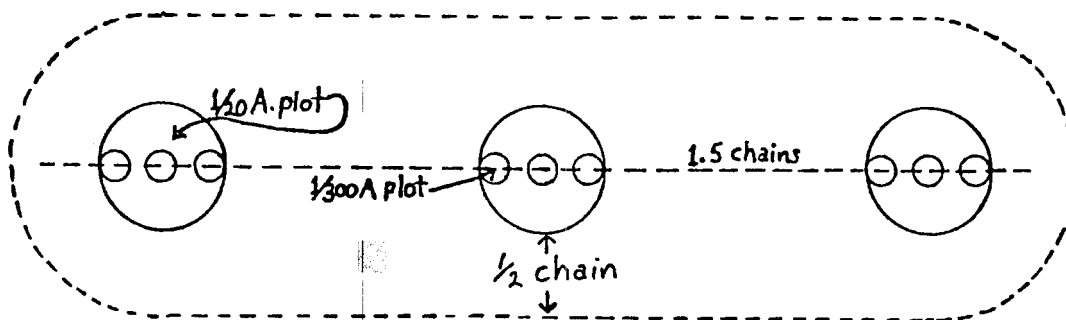
Whatever plot size is selected will be used for all clusters in the stand. Breakpoint dbh may be varied between control and treated clusters to avoid too many trees in the control cluster.

As a minimum, four clusters of the design shown in figure 1 will be used in each stand. If at least 40 trees 3 inches dbh and larger are not measured per cluster, additional clusters must be added until the total number of measured trees is at least 160. Some rules can be followed with a little common sense. This happens to be one of them. Suppose plot clusters were being installed in a stand just prior to an early precommercial thinning. Suppose there were relatively few trees larger than 3 inches dbh, but quite a few

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between 2 and 3 inches. It might make sense to consider 160 trees larger than 2.5 inches, or even 2 inches, sufficient. The breakpoint dbh would be reduced to 2.5 or 2 inches and trees down to this size would be measured on the 1/20 acre plots. On the other hand, suppose commercial thinning were being considered in a stand with an average dbh of about 8 inches, which also has an understory component with average dbh of 2.5 inches. The understory will be considerably knocked about by the thinning and finished off by clearcutting and site preparation 30 years later. What we want to estimate is the growth response on crop trees left after thinning. Therefore, enough plot clusters will need to be installed to pick up an adequate sample of such trees. This might be six or seven clusters. In such a case, use of larger plots would make more clusters unnecessary and would be a preferable course of action because fewer plots would have to be established to get the required number of trees.

623 - Treatment Vs. Control Plots. One objective of permanent plot installation is to assess the effects of silvicultural treatment. Therefore, it is desirable to make a comparison between treatment and no treatment, with as many other factors as possible held constant. The way to do this is to include in each plot installation one cluster which is not treated with the rest of the stand. This is the control cluster. It should be randomly selected so as to avoid, on the average, differences other than the treatment applied between the control cluster and the treated clusters. The control cluster should always be designated as cluster 1 with survey type 50. The treated clusters can be numbered 1, 2, 3, etc., with survey type 60. A control cluster with buffer strip is shown below.



624 - Locating Plot Clusters.

624.1 - Cluster Configuration. The plot cluster shown schematically in figure 1 consists of three 1/20 acre plots and nine 1/300 acre plots. Centers of the 1/20 acre plots are spaced 1.5 chains apart on a line. Centers of the 1/300 acre plots are located at the centers of the 1/20 acre plots and on the line 6.8 feet in from the edge of each 1/20 acre plot. Thus, each 1/20 acre

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plot will contain three 1/300 acre plots within it. The cluster center is defined to be the center of the middle plot in the cluster, unless the cluster has been rearranged to fit within stand boundaries. Trees of breakpoint dbh and larger will be measured on the 1/20 acre plot and trees smaller on the 1/300 acre plots. The three 1/300 acre subplots within each 1/20 acre plot constitute 1/100 acre per sampling point. Separate plot numbers are not used for each of the 1/300 acre plots. Therefore, record 100 for fixed plot size (Field 15) on the stand record sheet. Breakpoint dbh is recorded in Field 14. Generally, this will be 3 inches, but it need not be; nor need it be the same for treated clusters as for control clusters. After treatment, there may be only 20 to 30 trees on each 1/20 acre plot. These will be crop trees and there may be no smaller trees except for scattered seedlings. Then breakpoint dbh can be set at 0.2 inch and all trees taller than breast height measured on the 1/20 acre plots. Seedlings will be measured on the 1/300 acre plots. Breakpoint dbh should be not larger than 3 inches in any case. Where seedlings are especially dense, quadrants of the 1/300 acre plots may be used as described in 627.32.

624.2 - Stand Maps. Delineate the stand to be treated on both resource aerial photographs and topographic maps at approximately 1:12000--1:24000 scale. Further delineate the stand, if necessary, into general areas of homogeneous slope, aspect, elevation, and timber condition. Identify all roads, probable landing sites, and other areas known to be unavailable for treatment (stream protection zones, etc.). Select the homogeneous area in which you are more interested and outline it for easy reference on both the map and photo.

624.3 - Locating Cluster Centers. As shown in figure 2, place an X-Y coordinate grid over the stand delineated on the topographic map or aerial photo.

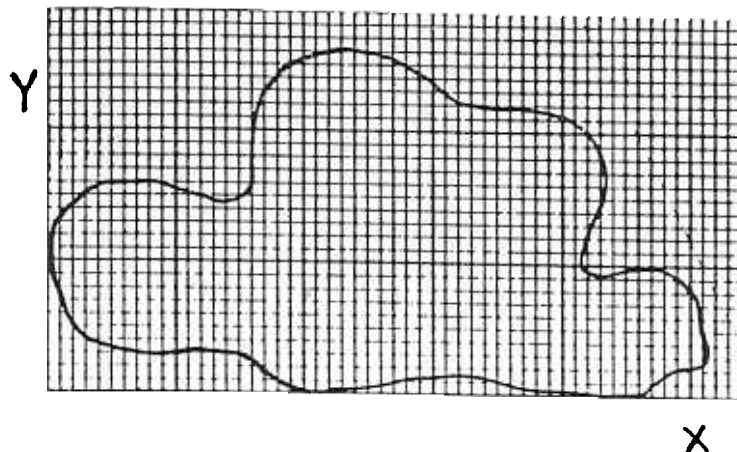


Figure 2. Locating cluster centers with X-Y coordinate grid.

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Orient the grid so that the (0,0) coordinate is in the lower left corner. Randomly select X-Y coordinates using tables of random numbers or the computer program R5*FS.RANDXY. An example of how this computer program is executed on a demand terminal and the results obtained is shown in figure 3. Locate four or more plot cluster centers within that portion of the stand which you want to monitor. The first cluster drawn will be the control cluster. It will not be treated. If any other cluster falls within one chain of the control, ignore it and draw another X-Y coordinate. Likewise, if a cluster should fall within an area you do not wish to sample, such as the rock outcrop or the stream protection zone, ignore that set of coordinates and draw another.

The azimuth of the line from plot to plot in a cluster need not be a cardinal direction; however, each cluster in the stand will have its axis laid out on the same azimuth. The azimuth shall be decided before the plot is located in the field. Determine the azimuth so that each cluster is as heterogeneous as possible. This is to say, run the clusters across skyline corridors, up and down hill, or across vegetative fingers.

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```
@XQT R5*FS.RANDXY
```

```
*** RANDOM X - Y GENERATOR ***
```

```
* XMIN, XMAX, YMIN, YMAX, NUMBERS (, SEED = 8 DIGIT INTEGER - OPTIONAL)
>0,240,0,210,50
```

```
MINIMUM X=      0      MAXIMUM X=      240
MINIMUM Y=      0      MAXIMUM Y=      210
NUMBERS  =      50      SEED    = 15573495
```

X	Y	X	Y	X	Y	X	Y
48	25	194	133	105	170	72	153
109	122	232	165	75	181	103	15
178	137	23	114	38	97	206	38
157	149	161	104	158	105	172	45
76	121	187	22	95	206	114	32
71	2	201	17	209	98	58	57
110	196	95	71	238	187	237	116
106	54	80	148	233	47	4	198
154	200	68	197	112	182	218	163
151	181	209	207	226	110	121	194
151	3	107	129	112	87	177	81
109	178	185	106	190	27	116	111
172	119	87	92				

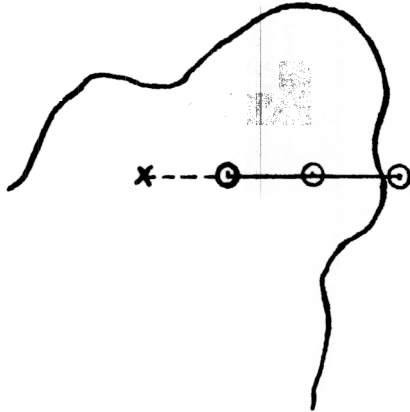
```
STOP
>
```

Figure 3. An example of executing R5*FS.RANDXY.

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624.4 - Keeping Clusters in the Stand. If a cluster falls near the stand boundary, be sure no plots fall wholly or partially outside the stand. At the risk of some minor bias, place the "outside" plot at the other end of the cluster, or at a right angle to the main cluster azimuth, so the cluster becomes L-shaped.

Example:



This cluster is partly outside the stand. Transfer the outside plot to where the X is. If that is outside, go 90 degrees one way or the other.

624.5 - Small Stands. If a stand is selected for a permanent plot installation, which because of small size or stringy shape does not allow placement of an adequate number of clusters, then twelve plots arranged in a systematic grid can be used. Then the plots should be grouped into four sets of three plots each, and each set of three plots assigned a common cluster number: 1, 2, 3, or 4. Randomly select one of these clusters to be the control cluster. Let it be cluster 1 and the others: 2, 3, and 4. All other information concerning data to collect, plot size, and documentation will be the same as for the cluster design used in larger stands.

625 - Equipment Needed for Plot Installation. The following suggested list of equipment is not exhaustive; additional items may be found useful, or the same tasks may be accomplished by other means:

1. Plot center stakes. The stakes should be of a material which will not deteriorate and of such strength they can be driven into the ground to a depth of a foot or more. Six inches to a foot of the stake should be left out of the ground. Sections of fiberglass fencepost work well. Two-foot lengths of galvanized conduit pipe (1/2" or 3/4") are permanent and rigid enough to drive. Conduit can be cut easily with a tubing cutter and is light enough to be easily carried into the woods. At least two types of stake material are unsatisfactory: flexible fiberglass and wire pins. The fiberglass is too limber to drive well and the wire pins are just no good in general.

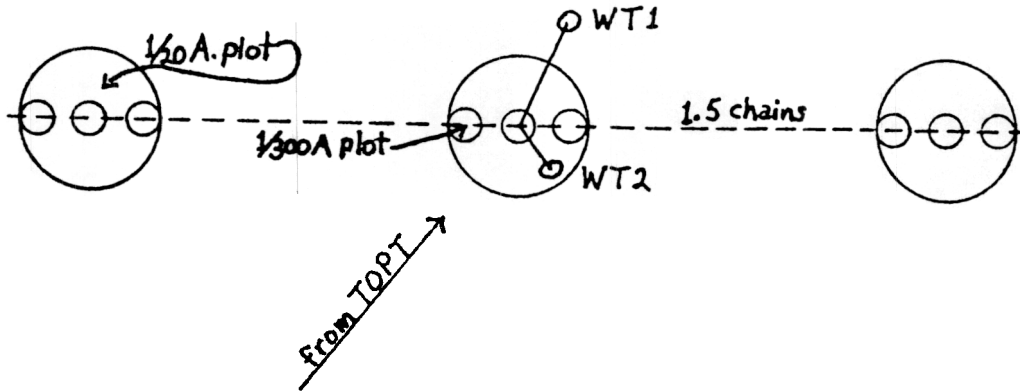
TIMBER MANAGEMENT DATA HANDBOOK

2. Five hundred numbered metal tags.
3. Aluminum nails at least 2 inches long. The longer the better.
4. Stand Examination Field Record Forms (R1-FS-2410-15B), one or more for each plot in a cluster.
5. Chain or 100-foot tape.
6. A 1/20-acre plot radius tape can be obtained for \$14.50 from Forestry Suppliers in Jackson, Mississippi. Use of such a tape is much more convenient than dragging the excessive portion of a 100-foot tape around the plot.
7. Metal number stamp for stamping plot number on fencepost. Not needed if tags are wired to post.
8. Tree paint in spray can.
9. Aerial photos of stand with cluster centers pricked and stand delineated.
10. Maps of vicinity and stand.
11. Pressboard folder with Kraft innerdivider GSA 7530-00-990-8884 for storing plot records.
12. Diameter tape.
13. Clinometer.
14. Tatum holder for plot record sheets.
15. Takeoff Point and Witness Tree tags.
16. A telescoping pole for measuring heights of trees in young stands.

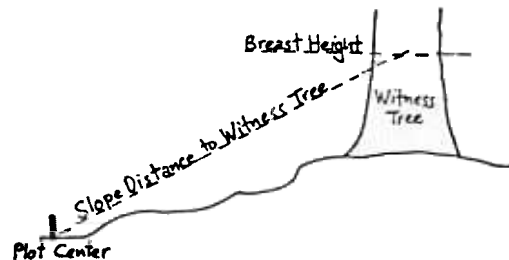
626 - Monumenting and Documenting Plot Location. Use takeoff point/witness tree (TOPT/WT) aluminum tags for referencing the plot clusters. A four-cluster installation will require at least 10 tags. Witness each cluster with two trees at approximately right angles to each other. In this way, if the plot stake is lost, it can be reestablished. If possible, try to have one witness tree

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facing in the direction from which the cluster will likely be approached.



Put a tag on each of these witness trees. Record in the notes the species, dbh, and bearing and distance from plot center to the witness tree. Distance is measured directly to tree center at breast height for these witness trees.



A takeoff point tree that references at least one plot cluster should be established at the point of departure from the road. The takeoff point tree should be easily spotted from the road, but not in a position that makes it likely to be removed by road improvement or reconstruction. Since it may be cut for some reason, a tag giving bearing and distance to the plot cluster should be placed on the tree below stump height. Another tag which will more easily attract attention if the tree is not cut should be placed on the takeoff tree above the level of surrounding brush. If low limbs are likely to hide the tag, prune them off. The takeoff tree and its location should be described in detail. Its species, dbh, and height should be recorded as well as the nature of surrounding trees. The position of the takeoff point tree relative to nearby topographic and manmade features should be described. A sketch map may be desirable. Be liberal in the use of paint and state in the notes what color was used. Flagging is of little use for any extended period of time.

626.1 - Takeoff Point Tree. The following example illustrates a proper takeoff point tree description:

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626.11 - Takeoff Point

Species: Engelmann spruce

DBH: 12.8 inches

Height: 87 feet

Marked with an unpainted aluminum tag at 5.5 feet above ground level and below stump height. Circled with a band of yellow paint below the upper tag.

626.12 - Relation to Plot

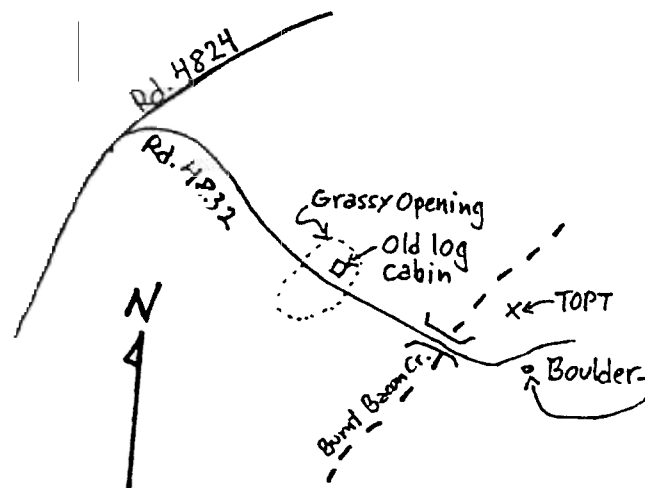
Bearing N. 37° E., 523 feet from the takeoff point to the center of plot cluster 1. At 203 feet, line passes 3 feet northwest of a very large cedar stump with springboard notches.

626.13 - Relation to Surroundings

The takeoff point tree (TOPT) is N. 62° E., 105 feet distant from the middle of the bridge where Road 4832 crosses Burnt Bacon Creek. From the TOPT, a large granite boulder on the south side of Road 4832 bears S. 12° E. about 70 feet distant. About 20 feet east of the TOPT is another spruce of similar size. About 50 feet west of the TOPT (toward the creek) are two grand firs 14"-16" dbh growing close together among grand firs and cedars 4"-6" dbh.

626.14 - Route to Takeoff Point Tree

Turn SE from Road 4824 onto Road 4832. Road 4832 is the one that crosses Burnt Bacon Creek and goes NE up the Mosquito Creek drainage for about 3 miles, where it crosses the ridge into the Sasquatch Creek drainage. Now, after turning from Road 4824 onto Road 4832, 0.45 miles from the road junction Road 4832 passes through a grassy opening with the remains of an old log cabin NE of the road. Burnt Bacon Creek is approximately 200 feet past the SE edge of this opening. (See sketch map below.)



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If this seems unusually detailed, consider that when this set of plots is to be remeasured, the Forest road-numbering system may have changed, or at least the sign for Road 4832 be torn down. The vicinity of the witness tree may have been logged, the road improved, the granite boulder shoved over the hill, and the bridge replaced. Relocating the plot will depend upon a few stumps, one of which it is to be hoped, will have an aluminum tag. Failure to relocate the plot or excessive time spent in relocation is extremely costly and can be prevented by good monumentation and documentation when the plots are installed.

Stand Description and Plot Measurement

627.1 - Stand Description. Use form R1-FS-2410-15BA to record characteristics of the whole stand. Some of these--slope, aspect, habitat type, and topographic site--will be recorded again for each point on the tree data record form. Diagram the stand in the place provided on form 15BA. The back of the form is blank and remarks can be continued there.

627.2 - Placement of Plot Centers. When placing plot center stakes, care should be taken to align the stakes accurately. This will help in relocating the plot center if a stake should somehow be misplaced. One way this can be done is to locate a preliminary center for the 1/20 acre plot. This is also the center of one of the small tree plots. Then, using a compass and tape, locate centers of the small tree plots which are tangent to the circumference of the large plot. These plot centers are to be aligned on an axis which is coincident with the cluster axis. After the small plot centers at the edge of the large plot are placed, move the preliminary large plot center in a direction perpendicular to the cluster axis until it is exactly on line between the two peripheral subplot centers. If use of the compass was accurate in placing the peripheral subplot centers, no more than a few inches of movement should be required to align the large plot center. Then the plot center stake can be firmly driven into the ground in its permanent location. The correct position for the large plot center stake can be judged by visual alignment between the two peripheral stakes, or by stretching a tape between the two peripheral stakes. When the plot is densely occupied by trees, these procedures are admittedly difficult to follow; nevertheless, a real effort should be made to align the stakes as closely as possible without cutting trees.

627.3 - Plot Measurement. After locating plot centers, including those of the 1/300-acre plots, determine which trees are in the plot. Place an aluminum nail in the tree at breast height on the uphill side. Use of a 4.5-foot stick to locate breast height will be helpful and is encouraged. Place a numbered tag on the tree below stump height facing plot center. Nail and tag all trees of

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breakpoint dbh or larger that are likely to be left after stand treatment. For control plots or treated plots measured after thinning, this means all trees of breakpoint dbh or larger. Taking bearings from the plot center, start tagging and numbering with the first tree clockwise from true north and proceed clockwise around the plot.

To record tree information on permanent plots, use form R1-FS-2410-15B. This form contains fields 34 and 35 in which azimuth and distance from plot center to each tree on the plot should be recorded. Do not use form R1-FS-2410-15BB for permanent plots.

627.31 - Large Trees. For trees of breakpoint dbh and larger, record and measure all stand exam information except age, height, and diameter increment for all trees. Record age and 10-year radial increment for the first tree in each of the following diameter classes for each species:

3"-4.9"
5"-8.9"
9"-12.9"
13"-16.9"
17"-20.9"
20.9" +

Never bore trees smaller than 3.0 inches dbh regardless of the breakpoint dbh specified. Do not bore trees in the 3.0"-4.9" class if it seems likely that doing so will retard their growth or survival. The object of permanent plots is to measure growth by successive measurements. If taking increment cores will interfere with this, we will forgo the cores. However, if a tree is bored, the core should be replaced in the tree.

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 Batch No. 16
 Page 1 of 11

USDA—Forest Service FIELD RECORD (Ref. FSH2409.21)		Forest Inventory Stand Examination		CARD 1 : BASIC STAND DATA									
STAND ID	Card Col. Desc Code Fields	1-2 Forest <u>07</u> F-01	3-4 Survey Type <u>60</u> F-02	5 District <u>1</u> F-03	6-7 Comp <u>17</u> F-04	8-9 Subcomp. <u>03</u> F-05	10-12 Stand <u>007</u> F-06						
BASIC STAND DATA	Card Col. Desc Code Fields	13 State <u>53</u> F-08	23-24 County <u>065</u> F-09	28-29 Strata <u>LVA</u> F-10	30-35 Date <u>10/19/85</u> F-11	36-41 Mo Year <u>10 1985</u> F-12	42-43 BAF <u>23</u> F-13	44-46 Breakpoint DBH <u>002</u> F-14	47-49 Fixed Plot Size <u>100</u> F-15	50-51 Radial Increment Growth Period <u>10</u> F-16	52-54 No. Sample Points <u>009</u> F-17		
55-56 Stand Slope <u>11</u> F-18	57-58 Stand Aspect <u>SE</u> F-19	59-61 Stand Elev. <u>043</u> F-20	62-64 Stand Habitat <u>693</u> F-21	65 Stand Topo Site <u>4</u> F-22	66-67 Stand Structure /Logged <u>12</u> F-23	68-69 Stand Stocking % <u>99</u> F-24	70-72 Stand Acres <u>037</u> F-25	73-74 D-1.0 in FINE DATA Sampling Planes F-26	75-76 1-3 in F-27	77-79 >3 in F-28			

OBSERVATIONS & COMMENTS; STAND DIAGRAM, PLOT GRID:

Route to stand: From State Highway 201, go north on Forest Road 4090 for 4.2 miles. Turn northeast on Forest Road 4093. At 1.8 miles from its junction with Road 4090, Road 4093 crosses a ridge. The TOPT is 46 ft. NW of the road centerline. The TOPT is a 16.3 in. larch with a tag below stump height and another 5 ft. above the ground. The TOPT has a band of yellow paint above and below the tag. About 20 ft. NW of the TOPT is a 12 inch Douglas-Fir, also painted.

Bearings and distances are as follow:

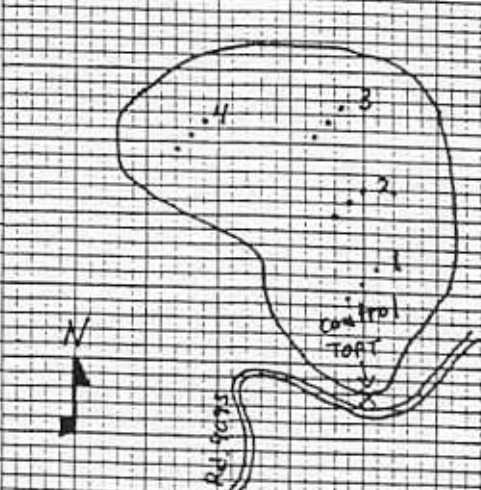
From	To	Bearing	Distance
TOPT	Center, Cluster 1	N 3° W	10 ch.
Center, Cluster 1	Center, Cluster 2	N 10° W	6 ch. 30 K.
Center, Cluster 2	Center, Cluster 3	N 16° W	7 ch. 25 K.
Center, Cluster 3	Center, Cluster 4	S 80° W	12 ch. 50 K.

Cluster #1 is the control cluster and was left unthinned. The rest of the stand was thinned to a 10 ft. x 10 ft. spacing in the summer of 1985.

3/4 inch conduit pipe was used to mark plot centers. Tops of the stakes are painted orange. Any rocks within 1 ft. of a stake were sprayed with orange paint.

The centerline of plot clusters was laid out on an azimuth of 45°.

Diagrams of witness trees at cluster centers are on the back of this page.



Examined 10/2/85 Edited 10/9/85 R1-FS-2410-15BA (1/83)

Figure 4. Form R1-FS-2410-15BA; to be used to record stand level information.

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R1-EDIT SYSTEM MASTER FILE FORMAT & UPDATE FORM										Batch No. <u>16</u> Page <u>2</u> of <u>11</u>																																								
USDA--Forest Service FIELD RECORD (Ref. FSH2409.21)				Forest Inventory Stand Examination				STAND ID																																										
BASIC STAND DATA		13	Card Col. Desc Code Fields	15	Action Code	23-24	State Code	25-27	County Code	28-29	F-10	30-35	Strata	36-37	38-41	Date	42-43	BAF	44-46	Breakpoint DBH	5	District	6-7	Comp	8-9	Subcomp	10-12	Stand																						
55-56	Stand Slope	11	5E	57-58	Stand Aspect	1	59-61	Stand Elev.	043	F-20	62-64	Stand Habitat	692	F-21	65	Stand Topo Site	4	F-22	66-67	Stand Structure /Logged	12	F-23	68-69	Stand Stocking %	99	F-24	70-72	Stand Acres	037	F-25																				
CARD NO. 1		CARD NO. 2																																																
LINE ID		Tree D No.		Sample Point No.		Action Code		Transact No.		Tree Class		CUT LV		Tree Species		DBH		Radial Growth or 5-Yr. Height Growth		Total Age		Tree Height Total/Truncated		Crown Ratio		Crown Class B-C-C		Damage		Cover Class		Record Once Per Sample Point																		
Card No. 2		Tree D No.		Sample Point No.		Action Code		Transact No.		Tree Class		CUT LV		Tree Species		DBH		Radial Growth or 5-Yr. Height Growth		Total Age		Tree Height Total/Truncated		Crown Ratio		Crown Class B-C-C		Damage		Cover Class		Record Once Per Sample Point																		
13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58					
2	12	02	01	001	-	009	212	01	1	-	1	052	31	025	047	4	2	45	45	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74			
2	12	02	01	002	-	017	052	01	2	-	-	AF	021	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
2	12	02	01	003	-	035	171	01	1	-	-	DF	038	22	021	036	7	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	12	02	01	004	-	053	104	01	1	-	-	L	046	28	024	040	5	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
2	12	02	01	005	-	071	197	01	1	-	-	AF	031	24	019	027	8	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2	12	02	01	006	-	087	013	01	1	-	-	DF	042	26	021	034	7	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2	12	02	01	007	-	101	116	01	2	-	-	DF	039	-	-	023	6	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2	12	02	01	008	-	118	221	01	1	-	-	L	044	-	-	032	6	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2	12	02	01	009	-	137	094	01	2	-	-	DF	039	-	-	030	7	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2	12	02	01	010	-	155	137	01	1	-	-	DF	041	-	-	032	7	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
2	12	02	01	011	-	173	143	01	1	-	-	L	048	-	-	036	7	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		

Card No. 1

Card No. 2

Crew E. Riese Date 10/2/85

Continued on Reverse

R1-FS-2410-15B (1/83)

G.P. Motsinger
T.J. Brooks

Figure 5. Front of form R1-FS-2410-15B; to be used to record stand (fields 1-25), tree (fields 30-52), and plot (fields 53-58) data.

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[illegible]

Figure 5a. Back of form R1-FS-2410-15B; to be used to record stand (fields 1-6), tree (fields 30-52), and plot (fields 53-58) data.

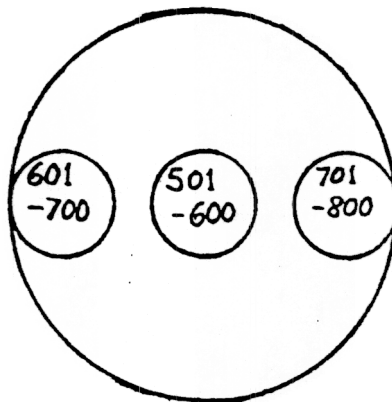
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Record heights on all trees on at least one-half of the plot, unless a tree has an unusual height for its diameter. Heights should be measured and recorded so as to provide an accurate height/diameter curve for the plot. This means the height sample should not only contain at least 15 trees, but it should cover the range of diameters on the plot. If the plot doesn't contain many trees in the first place, it may be necessary to measure heights on all of them. For measuring heights of 25 to 30 feet or less, use of a measuring pole is much preferred to use of a clinometer. Telescoping fiberglass height poles are available from Forestry Suppliers and the Ben Meadows Company and their use will result in more accurate height measurements.

In use, the pole should be positioned so the nail in the tree at breast height is level with the 4-1/2 foot mark on the pole. If there is no such mark, make one.

Large trees cannot have numbers attached to them which are greater than 500. After measurement, paint spot all trees to avoid duplicate recording. In some stands which have been thinned at a small average diameter, it may be desirable to set breakpoint dbh less than 3".

627.32 - Small Trees. Small tree subplots will be kept track of by the tree numbers: one subplot will have tree numbers 501-600, one will have numbers 601-700, and one will have numbers 701-800. These numbers are for recording only, as small trees are not tagged on the 1/300-acre subplots. They will be tagged when recorded on a 1/20-acre plot as described below. On a sketch map on the field sheet, indicate which numbers are associated with which subplot. An example is shown below.



For trees smaller than the breakpoint dbh, follow the guidelines in chapter 400 for small trees, with the following exceptions:

1. Do not code trees between 0 and .49" dbh as 001. Record the actual diameter. If the tree is of less than breast height, record 001 in field 40.

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2. Measure the azimuth and distance to only the four best small trees on each subplot. Record in fields 34 and 35.

3. Count all trees, whether they can be considered established seedlings or not. The reason for this is that judgment of the established condition varies between individuals and varies for an individual between plots. Thus, if only so-called established seedlings were counted, new regeneration could not be distinguished from what might have been on the plot previously without being counted.

Counting all seedling-sapling size trees will, in some stands, lead to an excessive number of trees on a 1/300 acre plot. In such a case, quadrants of the 1/300 acre plot can be used so that regeneration is counted on only 1/4, 1/2, or 3/4 of the 1/300 acre plots. However, to do this will require strict adherence to the following rules:

a. When the number of plot quadrants to be used has been decided upon, that number will be used for each and every 1/300 acre plot in the stand, except that the number may differ between control clusters and treated clusters. For example, one might choose to count trees in only two quadrants in the control cluster, but to use the full 1/300 acre plots in treated clusters. Then trees would be counted on two--and only two--quadrants of each of the nine 1/300 acre plots in the control cluster. The full 1/300 acre plot would be taken in all control clusters.

b. The decision on the number of quadrants to use must be made before any plots are taken; and once made, must be adhered to. The intent is to count about 15 trees per 1/300 acre plot. If regeneration density is not sufficient to attain this number on 3/4 or less of the plot, take the full 1/300 acre plot.

c. There will be no choice as to which quadrants will be used. If trees are to be counted on only one quadrant, it will be the northeast quadrant in all cases. If two quadrants are to be used, they will be the northeast and the southeast; if three, the northeast, southeast and southwest.

d. The number of quadrants used to sample small trees will change the area of fixed area plots associated with each of the numbered (1/20 acre) plots. If full plots are used, 100 is entered in field 15 for reciprocal of fixed plot size. If fewer than four quadrants are used (less than the full plot), then the following values are to be entered in field 15:

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Number of Quadrants Used	Entry in Field 15
1	400
2	200
3	133
4	100

Other standards of measurement and coding are the same as shown in Chapter 400 except:

Transect number = Cluster number

Point number = Plot number

BAF = 83 (for 1/20 acre plots, 84 (for 1/10 acre plots), or
85 (for 1/5 acre plots)

Fixed plot size = 100 (i.e., three 1/300-acre plots) or as
given above

627.33 - Small Trees in Seedling/Sapling Stands. Sometimes plots are installed in stands where the trees have no dbh or the dbh is not big enough to hold a nail. Usually stands of this nature have numerous small trees per acre. In such stands:

1. Install the control plot as previously instructed. This means some control plots will only have trees on the 1/300 acre subplots.

2. In the treatment plots:

a. Install the plot center stake and subplot stakes.

b. Record the tree information as previously instructed, except don't tag any small trees. Bearing and distance need not be recorded for more than the four best small trees. Small in this context means trees too small to hold a nail or which would be excessively damaged by a nail.

c. After treatment, change the breakpoint diameter to "002." That is to say, all trees taller than breast height will be on the 1/20-acre plot.

d. If the tree isn't big enough to hold a nail, make a wire loop around the tree and attach the tree tag to the loop. Be sure the wire will not constrict tree growth prior to the next measurement 5 years hence.

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e. Be sure to document in the notes all of these special instructions and anything unique that you do. Trees will receive a tag and paint spot at the post treatment measurement. A Supervisor's Office representative should check with Districts using these supplemental instructions for small trees to make sure the stand is of a type needing these special instructions.

628 - Documentation. Keep the records for the stand in a pressboard folder (GSA 7530-00-990-8884). In the folder will be aerial photos, maps, plot sheets, and directions on how to get to the stand and clusters. The folder will also contain information on general stand conditions and a copy of the silvicultural prescription. Note information on date of treatment, who did the work, and with what equipment.

After the tree and plot data are entered in the R1-EDIT system (placed on a permanent or temporary master file), run a set of stand tables for the set of permanent plots and include them in the plot folders. A stereo pair of photos covering the stand in which permanent plots are located must also be included in the folder for each plot installation. The stand boundary should be shown on the photos and cluster centers should be pinpricked. The cluster number associated with each pinprick should be written in ink on the back of the photo.

When entering permanent plot data into the R1-EDIT system, the first measurement on treated clusters (following treatment) is coded as survey type 60. The first remeasurement is entered as a survey type 61. The control plot cluster is entered as a survey type 50. The first remeasurement of control plot clusters will be coded survey type 51. Accomplishment reports are generated from the timber stand data base. Therefore, to get credit for installing permanent plots in a stand, the stand will have to be in the data base with activity code 4036 (Permanent Plot Installation) and the appropriate accomplishment year. In order to be included in the fiscal year accomplishment report, permanent plot measurements or remeasurements must be recorded in TSMRS by September 30 of each year. Subsequent remeasurements will be coded as survey types 62, 63, etc., for treated clusters and 52, 53, etc., for control clusters. The TSMRS activity code for permanent plot remeasurement is 4037.

A file of permanent plot folders will be kept with the Forest coordinator for permanent growth plots. A copy of each folder will be sent to the R0 Timber Management Plans and Programs group. Plot records will be filed in a pressboard folder with Kraft inner divider (GSA 7530-00-990-8884). Because of limited storage space, use of this folder is essential. Any permanent plot records sent to the R0 in any other form will be returned to the Forest to be put into the correct folder. This folder will be sent to the R0

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via the Forest Coordinator who will review the adequacy of documentation before transmittal to the RO. It is important that all the items mentioned above be in the folder. Plot folders from newly established plots are due in the RO by February 1.

629 - Permanent Plot Remeasurement

1. When a permanent plot installation is selected for remeasurement, the first thing to do is find the plots. Get the plot folder from the files and look at the plot location on the photos. Carefully read the installing crew's description of the route to the stand and the markings at the takeoff point. Layout the route on a map and proceed to the takeoff point.
2. When the takeoff point is located, refresh all paint marks. Replace the takeoff point tree tag if necessary. If the tag is alright, but growth of the trees appears likely to push the tag off the nails before the next measurement, the nails should be pulled out with a claw hammer until just a portion of each nail sufficient to hold the tag remains in the tree.
3. When the cluster center and plot centers are relocated, pull nails or replace witness tree tags as described above. Refresh all paint. This should not be done until after the trees are measured, for obvious reasons.
4. As a first step in remeasurement of a plot, all sample trees should be located. When a tree has died, it should nevertheless be measured and coded as a dead tree. On live trees, the nail at breast height and the nail holding the tag at the stump will be pulled enough to prevent overgrowing. Occasionally, a tag will become lost from a tree. It is best if it can be replaced by another tag bearing the same number as it had originally. Often that will not be feasible because no tag with that number will be available. Then, use any tag as long as the number does not duplicate that of another tree on the same plot. When tree numbers are changed in this way, the tree number used at the time of previous measurements should be changed on plot record sheets and on the REDIT master tape so as to agree with the tag currently on the tree.
5. All measurements of dbh will be made at the point on the tree stem marked by the nail. This is where the previous measurement was taken and only by measuring at the same place on the second occasion can dbh growth be obtained. If the nail was not correctly placed at breast height during plot installation, dbh is still measured at the point indicated by the nail.

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6. Occasionally, height of a tree may appear to be less at remeasurement than it was at the first measurement. If this happens and it cannot be accounted for by visible top damage, remeasure the tree again from a different location. Perhaps the first attempt at remeasurement was in error. But if the most careful remeasurement is still shorter than the previous height recorded for that tree, record the accurate remeasurement. Do not change the previously recorded height, although it is obviously incorrect, unless internodes can be counted accurately and height at the previous occasion can be accurately measured. If the tree has suffered top damage, record the appropriate codes in Fields 46, 47, and 48 (see FSH 2409.21h, R-1 415.4).

7. Breakpoint dbh will not be changed from what was established at the time of plot installation. There may be trees which were smaller than breakpoint dbh at the time of previous measurements and were either measured on one of the 1/300 acre subplots or were not measured at all, but have since grown to be larger than breakpoint dbh. These are called ingrowth trees. They should be assigned a tree number which is different from the number of any other tree on the plot. Affix a tag bearing the tree number below stump height and drive a nail at breast height. Measure ingrowth trees in the same way that all other trees are measured on the larger plot and enter the measurements on the plot record sheet. Enter the numeral 4 in Field 59 for ingrowth trees the first time they are measured. They will not be recorded as ingrowth trees at subsequent remeasurements. There may be other newly ingrown trees at that time. Usually, numbering of ingrowth trees will take up where numbering of previously measured trees left off. For example, if the highest numbered tree at the previous measurement was 70, numbering of new ingrowth trees will begin with 71. Numbers of trees which disappear from the plot because of mortality or removal will not be reassigned to new ingrowth trees.

8. Occasionally, remeasurement will discover a tree on the plot which was missed and not measured at the previous occasion. If such a tree is encountered and it is ascertained it was not, in fact, ingrowth, then it will have to be assigned a number as described above for the case of missing tags. The tree will be measured and recorded on the plot record sheet. It will also be recorded on the plot record sheet(s) for previous measurements. Corresponding insertions will be made in R-1 EDIT master tape records. A dbh for the previous measurement will be calculated as

$$dbh_1 = \frac{dib_1}{dib_2} dbh_2$$

where dib_1 and dbh_1 are dimensions at the previous measurement and dib_2 , dbh_2 are current measurements. Diameters inside bark

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are obtained from an increment core. If height of the tree at the time of previous measurement can be established by counting internodes back from the tip and measuring height to the appropriate whorl with a clinometer or pole, then that is the appropriate procedure. Otherwise, do the following:

a. Select a sample of trees on the plot (or on the cluster, if necessary) having approximately the same dbh as the newly discovered "in" tree. This sample should consist of at least six trees.

b. Calculate average height ($\overline{H2}$) and average height at the previous measurement ($H1$) of the sample trees.

c. Backdated height ($H1$) of the previously missed tree is then calculated as):

$$H1 = \frac{\overline{H1}}{\overline{H2}} H2$$

Before assuming a tree not recorded in the previous measurement is a missed tree, one should make sure it is not ingrowth which was smaller than breakpoint dbh at the time of previous measurement. To ascertain this may require boring the tree. Boring should be avoided, if possible, but if it is necessary, it must be done.

9. No trees will be bored at remeasurement except to establish whether the tree was ingrowth or a missed tree. Specifically, no growth sample trees will be bored. Also, no trees will be sampled for height growth.

10. Plot record sheets for the remeasurement are to be entered in the plot folder. Copies will be sent to the RO for inclusion in the folders kept there. Remeasurement data are to be entered into the R1-EDIT system with survey type code 51 for control plots and 61 for thinned plots. The second remeasurement will be coded 52 and 62, etc.

11. If more recent aerial photography of the stand containing the plot installation is available, obtain a stereo pair of prints covering the stand and include them in the plot folder after transferring all pinpricked points to the new photos. The old photos should be retained in the folder.

To: Permanent Growth Plot Coordinators and Stand Exam Crew Supervisors
From: Jim Brickell and Tina Naugle (FTS 585 329-3280)

4/13/92

Some Answers to Questions About Permanent Growth Plots

Many people are already heading out to remeasure permanent growth plots and others may soon be taking a look at their RLEDIT Master Lists and TMSTAND #43 Permanent Growth Plot Status Reports. We are sending the following information which we hope will help clarify some questions people have raised which will ultimately be addressed in an Amendment to Chapter 600 of the Timber Management Data Handbook. Even if you have done permanent growth plots before, in addition to reading this document, be sure to review Chapter 600 which should be referred to often in the field by everyone on the crew to insure consistent data collection!

1 What exactly do we send to the RO?

For every growth plot that has been established, there should be a folder at the RO. The documentation we want in the folders at the RO is described in Sec. 628 of the TM Data Handbook. If the folder is not here, we will first check at the SO/Zone level and then get in touch with the district. If you have folders made up for the RO that haven't been sent in yet, send them to Jim Brickell at the RO (by USPS, UPS or hand-carried-- see schedule and address at end).

If you have extra copies of remeasurement data (in the form of original plot sheets or an RLEDIT Stand Listing) and RLEDIT tables which you haven't sent to the RO yet, send them, along with a brief write-up about the remeasurement (include any difficulties encountered, general stand condition, description of remonumenting activity, etc.). If you don't have copies of remeasurement data and tables already made for the RO folder, just send the brief write-up. From time to time, we may be requesting that you send copies of the original data sheets if we are unable to reconcile discrepancies using the RLEDIT data. The better edited the data are on your end, the less questioning we will have to do on this end!

If updates have been made to original data files since you sent in their paper copies, please send a note indicating that, or, if you have them, new copies of the stand listing and tables. The first thing we check are the stand folders -- printouts or xeroxes are a lot easier to look at than the DG screen. If we know that the RLEDIT files have been updated, then we can extract them and make a new copy. If you have already printed out the post-update stand listing and tables, it would save time and paper if you send what you have printed.

If you haven't yet entered data from growth plots which have been accomplished, do so, and make sure they are well-edited before you send them to Kansas City. That means look at the pre-edit stand listing, and make sure there are not trees with diameters of 0, or heights of 90 instead of 09, etc. You will find on remeasurements that the missing values table at the end will have many "still missing values" for age and radial growth on trees over 3" DBH because you will not be drilling them again (that's ok, we already know they are x years older than the last measurement and will get growth data from the difference in diameters and heights over time). The analysis program will use the raw growth plot data, not RLEDIT's fill-in values.

In addition to the above, it is a good idea to keep a summary at your District and Forest of the stands with growth plots, noting any idiosyncracies, e.g., about plot design, measurement periods or treatments.

2. What do we do if a cluster or part of a cluster has been damaged or destroyed?

If it is a case where the control cluster was thinned due to lack of communication with the thinning crew, there may be no opportunity to establish another control plot because the rest of the stand may have been entirely treated, or any untreated area which remains is not big enough to accomodate another cluster. If a good effort to find an untreated area of the original stand with the same site conditions fails, designate the densest of the thinned clusters as the control cluster, with a survey type of 50. Don't lump the former control cluster with the other treated clusters under a survey type 60; if you have already done this, you will need to look at the plots on the ground and then decide which cluster to call the control.

If the destruction resulted from running machinery or skidding logs through the cluster, and more than one plot in the cluster is destroyed, then a new control cluster will have to be established in the densest part of the stand that can be found (even though it may have been treated), with the survey date being the time of installation. This may mean that we've lost one growth period, but there's probably no help for it.

When only one plot of a control cluster has been destroyed or disturbed and the remaining two plots are still fairly undisturbed, it may be possible to establish a new plot somewhere within the untreated buffer strip surrounding the control cluster. This would be preferable to replacing the entire cluster.

In that case, backdate the measurements to the time of measurement of the two original plots. Try not to do this by boring the trees. Using data taken from the other two plots to plot growth over dbh (separately by species), draw a curve through the data points, and use that curve to estimate an amount to be deducted from the present dbh of trees on the new plot in order to backdate their size to the date of the previous measurement. Do the same to backdate present height measurements for those trees where whorl counting is impractical. If whorls can be counted on most trees, backdate height by that means and consult the RO on a statistical procedure to use dbh and height growth to backdate dbh.

If the destroyed cluster was a treated one, you will have to relocate the plot(s) and backdate the measurements to the time of measurement of the other treated clusters so the survey type will be the same for all the treated clusters.

Needless to say, effective communication and contract administration can go a long way in preventing mishaps of this sort. Keep in touch with crews and contractors when you know they're in the vicinity of your permanent growth plots, and if you don't know, find out if they might be!

3. How do we number "best" trees that were on the small (1/300 acre) plots and have now ingrown to the large (usually 1/20 acre) fixed plots?

→ Keep the original 500-700+ numbers throughout the measurement series.
New tags can be made by stamping the numbers on blank tags. This is a much

better practice than the use of prestamped tags, which are usually purchased in groups numbered from 1 to 100, 101-200, etc. They rarely all get used on a particular plot, and to avoid tag wastage, they sometimes get used out of order on another plot for replacing lost tags and tagging ingrowth trees.

When replacing a tag is necessary, the same numbered tag is often not available, so a different number is used instead. Doing this requires changing the tree number in previous measurements so that identity of the tree is not lost. But that often doesn't get done, and even if it is done, the field sheets in all locations may not get changed after the update is made.

Stamping a blank tag may take up to a minute of time and require carrying a set of ten numeric dies to the field, but they don't cost much and don't weigh much and in the end this may cost less than accumulation of prestamped tags that cannot reasonably be used. The desire not to waste prenumbered tags has led, in at least one case, to tag numbers on trees which are different from the tree numbers recorded on the field record sheet. That is confusion compounded.

If you keep your leftover tags well-organized and identified, they can be used on the next remeasurement for ingrowth trees so that numbering remains sequential with the previously measured trees. And of course, you can always recycle leftover or used tags and nails.

4 What if the ingrowth tree was part of a group on the 1/300 acre plot?

If the tree which grew larger than breakpoint dbh was originally part of a group on the small plot, with some trees of the original group remaining below breakpoint dbh, then it will be necessary to assign the tree a new number inasmuch as the same number cannot be used for both the remnant of the group and one of its former members which is now measured on the large plot. In that case assign the ingrowth tree a number larger by one than the highest number used for large plot trees heretofore. Or larger by two or three, etc., as necessary to accomodate all such trees that grew out of a group.

Of course some folks will want to do it the other way around, but as long as some trees of the group remain below breakpoint dbh it seems more reasonable to use the original group numbers for those trees which are still on the small plot. The 500-700+ number will then be retained for the last member of the group to grow past breakpoint dbh, if it ever does.

5. a. How do we code the ingrowth from the small (1/300 acre) plots to the large (1/20 acre) plots? b. What is the GST code 6 for?

a. With a GST code of 4 in field 49.

b. There is a GST code 6 (see Timber Management Data Handbook 415.54), but it is reserved for special cases where the breakpoint was decreased in order to have a larger sample of trees on the 1/20 acre plot; it indicates a tree which is now on the large plot that wasn't measured previously, but would have been above the new breakpoint in the previous measurement, thus differentiating it from ingrowth in the usual sense

6. How do we code trees which grew into the small plots (new regeneration)?

Give the tree or the group a new tree number in the 500-700+ series, depending upon which 1/300 acre plot it is on. They do not get a GST code for ingrowth.

7. Do we take GST's on the remeasurement?

For trees greater than 3" DBH, it is not necessary and not recommended, as increment boring may cause growth aberration in the sample trees. On smaller trees, count whorls and get height growth on 1 or 2 trees that are ingrowth (and not measured before). Occasionally, you may notice an age or growth recorded previously that appears to be an error. If you are suspicious, don't bore the tree! Instead, use whorl counts to check ages and the differences between dbh measurements to check the previous radial growth measurement. If the tree is too large for whorl counts, either give the previous crew credit for doing it right, or use your judgement to arrive at a more reasonable estimate.

8. Should there be 4 best trees on every little plot at every measurement?

Yes, if there are at least 4 trees on the plot, give an azimuth and distance (to the nearest 1/10 foot) to the four best ones, including new seedlings if that's all there are. If one becomes ingrowth to the large plot, select another best tree to replace it. The best trees are what you think may become crop trees, or are of a different species, such as spruce in an LP stand, which might be interesting to track through time. If you haven't been recording best trees on remeasurements, that's ok-- record them next time!

Other trees on the small plot can be grouped following FSH 2409.21h R-1 415.21 procedures if they are so numerous as to make recording them individually infeasible. There is no requirement to group small trees, and unless they are virtually identical, a dozen or so small trees can be recorded and tracked individually with little difficulty. But it is futile to assign, for example, 50 tree numbers that cannot possibly be tracked without distance and azimuth recorded.

If a new best tree comes from a previous grouping, assign the group number to the best tree and assign a brand new number to the remaining group (this may seem inconsistent with the instructions for leaving the group number with the group in the case of a non-best tree ingrowing to the big plot, but since we are interested mainly in crop tree growth, the best trees are the ones which will be tracked).

Obviously there is a need for good judgement with grouping: if you group too much without anticipating potential crop trees that could be kept separate, you will have to keep making new group numbers as trees ingrow and if you don't group at all, you will be spending a lot of time with unnecessary recording and data entry. Whatever you do, keep good notes, either in the margins of the plot sheets or in a field notebook, transferring them to a summary write-up in the permanent plot folder.

9. How should small trees (< 3 inch DBH) be tagged?

Tagging small trees can be difficult. Usually they are far too small to fasten a tag with a nail. A wire loop large enough to allow for growth around the bole may not be the best idea either because the wire usually ends up being held against the bole by gravity and it can be grown over. Wiring the tag to a branch is better, but even then it should be a branch that does not appear likely to contribute much to photosynthesis. A wire loop holding a tag and laid on the ground is not useful because animals are likely to disturb the tags, perhaps disturbing the tree in the process.

- 9 cont. Trees that are too small just cannot be reasonably tagged. To minimize confusion, if you cannot tag a tree, make sure the azimuth and distance from the plot center are accurate (on the 1/300 acre plots, it is helpful to measure the distance to the nearest 1/10 foot).
10. When there are three 1/300 acre plots per large plot, what is the fixed plot size on Card 1?
The correct code is 100. This is a common error on many districts-- think of the small fixed plot as a 1/100 acre plot associated with a large plot that happened to get split into 3 smaller plots of equal size.
11. How do we account for small fixed plots which don't have trees on them?
What do you do on stand exams? Write "no 500 fixed plot trees" and be grateful! Unless you are recording cover data on all of the small plots, do not write 501, 502, etc. with no tree data following. If there are no trees this time, there may be next time! Remember, if there are trees, they have the same cluster and plot number as the large plot.
12. Do we take cover data on all of the 300th acre plots?
You may, but it's not required. You are required to record cover data and site data (slope, aspect, habitat type and topo position) for the center 1/300 acre plot on each of the three large plots in a cluster (the data goes at the beginning of the plot record, just like in a stand exam). Continue to use the 2-digit cover type codes in fields 50-52 even if you are also filling out a Card 9 for ECODATA. (If the site data differ much from plot to plot or from cluster to cluster, the installation has not been very well located, but we'll have to live with it).
13. How many heights need to be measured on the large plots?
On the treated plots, at the very least 15 trees, but it won't hurt to measure more, especially if there are several species and a wide range of diameters. On the control cluster plots, at least 10 trees of intolerant species and 10 of tolerant species, covering the range of diameters, including any large relicts. If you record a height, make sure it is measured accurately or estimated from an adjacent tree not more than 8-12 feet apart.
14. A contracted crew used 600-800+ numbers instead of 500-700+ numbers...
Change them in the data files, and if there were ingrowth trees from the 1/300 acre plot, make sure it is well-documented, and change the tags at the next remeasurement.
15. One of our crews used a 1/20 acre plot radius corrected for slope instead of using horizontal distance (26.3 ft.)...
On the remeasurement, if a lot of trees would have to be deleted from and added to the previous measurement in order to change the plot configuration, leave it the way the crew established it and make sure it's documented well.
16. How should we deal with trees that are no longer standing in the post-treatment measurement (i.e., were thinned), if there was a pre-treatment measurement?
Just keep track of the trees remaining, including standing dead trees which were not intentionally cut or killed. To code the thinned trees as

than 100, but this should be done only if you are really driven to it, and only after you have discussed it with the permanent growth plot coordinator at your SO. Don't get carried away with the idea of raising the breakpoint dbh. Nobody wants to do a lot of work that isn't necessary, but don't get the idea that the objective is to follow only the initial set of trees through time. After a thinning there are sometimes enough new trees established to compete seriously with the anticipated crop trees. If you keep raising breakpoint dbh to only measure the trees that were measured on the first occasion, all of this additional competition will be missed. Those new trees are there, and they should be accounted for. Don't rely on the 1/300 acre plots to pick them up satisfactorily.

If you do change the breakpoint dbh, leave the previous breakpoint whatever it was. There is no point in deleting the newly excluded trees from the previous measurement. Some of the trees previously below the new breakpoint dbh will now be above it. By deleting their records from the previous measurement, you will throw away information on their growth.

When selecting a new breakpoint, use one that will keep almost all of the previously measured trees in the sample, i.e., don't raise it so much that you lose a lot of trees. Someone measured them all before; you can do it again. On the other hand, eventually, perhaps in 10 years or more, some of the present ingrowth may outstrip some of the original trees in growth so that keeping breakpoint dbh low enough to contain all of the original trees may lead to an absolutely excessive number of trees. If that actually does happen, then you may have to raise breakpoint dbh enough to exclude some of the original trees. If a few trees end up not being on the plot because they are below the new breakpoint, don't reuse their number for other trees, as they may grow into the plot by the next measurement.

19. a. What if we don't have the funding to do our growth plot remeasurements?
Talk to the SO, and if that doesn't help, call Jim Brickell at the RO.

- b. What if we cannot do it this year?

If the remeasurement has to be delayed a year or two, at least get out there, to make sure the plots are locatable, and pull the nails and wires. When pulling nails out, use a glove or small piece of wood as a fulcrum to avoid damaging the bark. If a tree has grown over a nail, do not try to remove it, same with a wire if the bark has grown over it. To avoid this occurrence in the future, make sure nails are at least 3 inches long and don't pound them in very far. Use aluminum tags and nails so if it does happen, possible future damage to saws will be minimized.

Approximate schedule for PGP cleanup

April: Kootenai, IPNF and Clearwater
May: Clearwater, Flathead and Nezperce
June: Nezperce, Lolo and Bitterroot
July: Lewis and Clark, Gallatin and Beaverhead
Aug.: Helena, Deerlodge and Custer

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